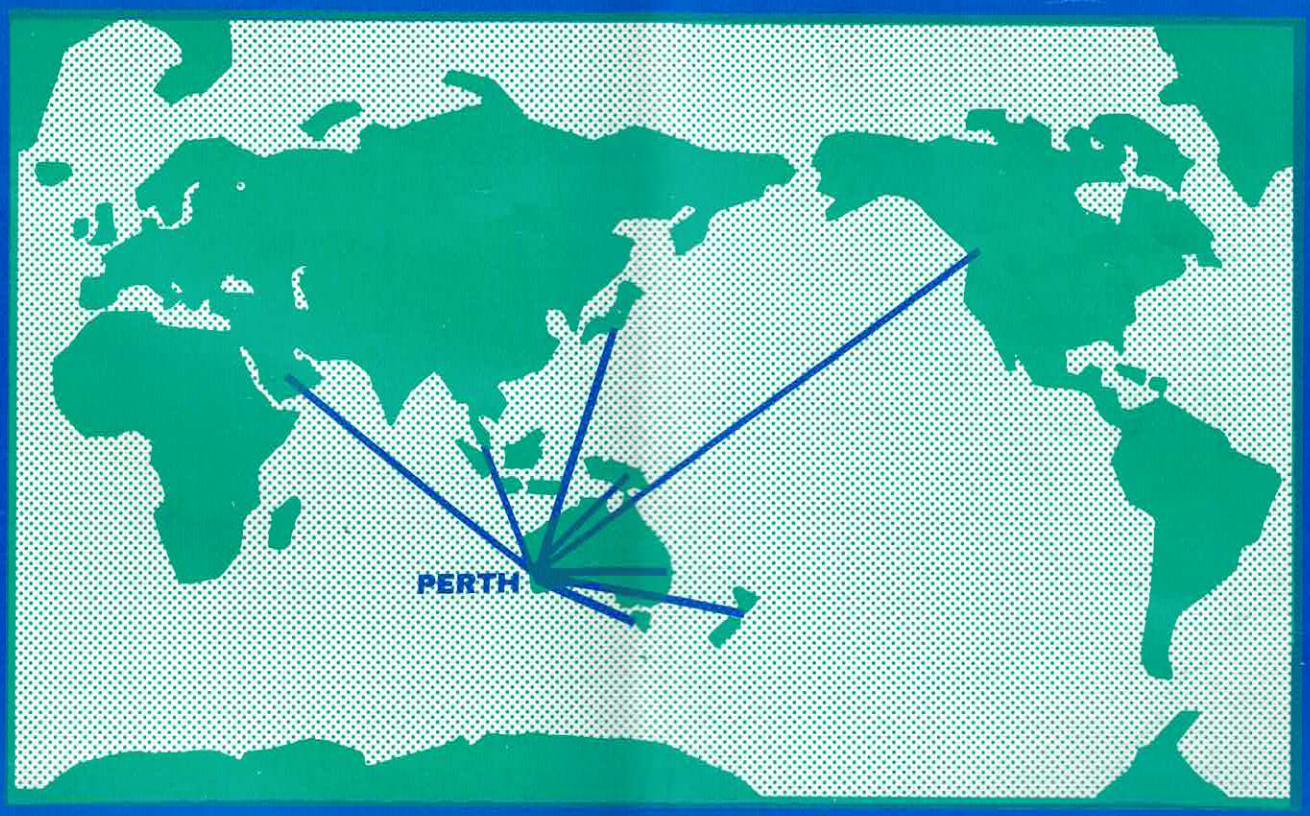
MSX and SPECTRAVIDEO

COMPUTER FORUM

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EDITORIAL

I thought I would start off this editorial by answering a few queries we have had, first off we leave various articles or columns out of the magazine because of our size, well make it bigger you may say!. We can't make our magazine much bigger because of the costs involved. Your next statement is 'This issue is bigger', yes it is and there is a good reason for this. We have left out the Adventure column and are running the same cover so we could use the money we normally pay to those people so we could have a bigger magazine.
Now the reason for the bigger magazine, it has been 12 months since Computer Forum was on the shelves of the Newsagents. This means there is more than 170 subscriptions due this issue and for our magazine to keep going we need those people to re-subscribe, so, if your subscription is due please send your form back as soon as possible.

You will also notice in this issue a new Column (Puzzle Corner) this column has been kindly donated by PAUL LEWIS of WA, and Paul hopes to keep his column going as long as possible and invites reader criticism or input.

We have received a press release from the ASCII Corporation in Japan. It explains how ASCII started and where they are today it is good reading to all you MSX'ers out there.

We are running very short of listings for MSX and SVI so if you have any programs no matter how big or small send them in and if we use them you will recieve payment for them, but please include return postage if you want your tape or disk back.

One more thing before I sign off, I have had a bit of a chat with some people about starting up a swap column for software, if you are interested in this idea please drop us a line and if we have enough interested people we will do one.

PUZZLE CORNER

PUZZLE CORNER will, each month contain a puzzle, changing form each week between; 1) WORD SEARCH
2) CROSS WORD
3) ANAGRAMS
4) ALPHAGRAMS
5) MISSING VOWELS

Plus any requests or examples sent in from you. If any QUERIES arise concerning certain words, answers or even questions, please do not hesitate to send them in...I LOVE CRITICISM. (I may even publish certain queries with my answer.)

AHHH I hear you say....ANSWERS.....these will be published in each following issue (only a months suspense). The puzzles are centred on computers, computer words and jargon and will hopefully become harder in each issue. A time limit will be placed on certain puzzles -- see if you can beat the clock as well.

G O O D L U C K

WORD SEARCH 1 (Time limit 10 mins)

B	U	F	F	E	R	X	E	D	O	C	R	A	B	Y	T	E
A	S	U	U	N	I	V	A	C	P	O	A	S	A	B	P	C
T	E	N	I	L	N	O	Q	E	T	T	E	K	S	I	D	O
C	R	D	V	A	B	F	R	S	N	O	B	B	I	T	L	R
H	E	L	P	V	U	F	I	L	E	B	N	L	C	I	N	E
P	A	R	I	T	Y	L	S	T	P	O	L	M	N	N	P	B
R	C	X	W	N	H	I	F	G	T	R	O	E	S	A	Z	Q
O	Y	A	A	C	N	N	A	T	H	O	B	L	E	M	E	L
C	Z	D	R	O	W	E	E	H	G	C	O	H	T	T	T	A
E	C	A	B	D	D	I	V	D	I	S	C	Q	Y	A	Y	T
S	E	Q	U	E	N	C	I	A	L	X	E	B	B	U	B	I
S	S	E	C	O	R	P	J	T	T	D	A	N	A	L	O	G
I	B	I	N	A	R	Y	M	A	X	G	E	S	G	Q	L	I
N	A	R	T	R	O	F	Q	K	E	B	U	U	I	E	I	D
G	E	R	M	A	N	I	U	M	T	S	L	B	G	X	K	L

The words listed below can be found within the block above in straight lines, either upwards, downwards, sidewards or diagonally. Oh and some words may be backwards in the above directions.

LIST OF WORDS CONTAINED

LINE
CHEAP
SEARCH LIST
CORE
BUG
BUFFER
BATCH PROCESSING
SEQUENCIAL
OFF LINE
ON LINE
BIT
LINE
BUS
BAUD
BYTE

DISC
BINARY
DATA
FILE
DISKETTE
DIGITAL
ANALOG
BASIC
COBOL
FORTRAN
UNIVAC
PARITY
GERMANIUM
ROBOT

HELP
CARD
PROCESS
LIGHT PEN
BAR CODE
KILOBYTE
MEGABYTE
GIGABYTE
XEQ
USER
FUN
VENN
WORD
TEXT



HIGH GAME SCORE

GAME	NAME	SCORE	LEVEL
ALI / FORTY TH	PETER COTTER	80,100	
ANTARCTIC ADV	DAMON BURN	67,660	STAGE 11
ANTY	JARED HYLAND	122,600	
ATHLETIC LAND	PETER COTTER	324,500	STAGE 25
BATTLESHIP CLAP	JOHN KNIGHT	60,100	
BEAM RIDER	RICHARD JOHNSON	81,112	
BUCK ROGERS	JASON VESPA	149,000	
CLAPTON 2	JASON VESPA	70,400	
COMPUT BILLIARD	PETER COTTER	114,300	
DUNGEON ADV-	PHILIP BROUWER	LEVEL 3	
DORODON	JOHN KNIGHT	45,540	
EXA INNOVA I.E	JOHN KNIGHT	25,710	
GOLF	DAMON BURN	5 UNDER	
GREEN BERET	PETER COTTER	116,300	
H.E.R.O.	JOHN KNIGHT	51,235	
HUMPHREY	KAYE BERGMAN	4,020	
HYPER SPORTS	P. QUINN	475,000	
KING VALLEY	PETER COTTER	49,300	
KNIGHT TYME	CRAIG HURSEY	95%	
LE-MANS	MATHEW HYLAND	35,250	
LODE RUNNER	PETER COTTER	279,575	
MANIC MINER	KEVIN TATE	66,663	
MONKEY ACADEMY	JOHN KNIGHT	127,490	
MOUSER	JOHN KNIGHT	35,900	
MR WONGS L/LAUN	JARED HYLAND	1,430	
PIN BALL	B. RANKIN	870,450	
PITFALL 2	P.J. CHRISTIE	199,000	
POLAR STAR	R. DUNCAN	210,000	
RAID BUNGELING	P.J. CHRISTIE	112,030	42 MINUTES
ROAD FIGHTER	CRAIG HURSEY	190,235	STAGE 14
RIVER RAID	D. OLSON	56,650	
SCION	JOHN KNIGHT	131,400	
SENJYO	JOHN KNIGHT	89,150	
SKRAMBLE	DEL BERGMAN	24,370	
SPECTRON	PHILIP BROUWER	9,640	
SUPER COBRA	JARED HYLAND	113,440	
SUPER CROSS FC.	JARED HYLAND	259,000	
SUPER GOLF	PETER COTTER	-1	UNDER PAR
SUPER TENNIS	DAMON BURN	6 - 0	
SWEET ACORN	JASON VESPA	174,380	
TENPIN	LEE BERGMAN	124	
TRACK & FIELD 1	PETER COTTER	704,630	
ZAXXON	JOHN KNIGHT	52,200	

Music Competition

We have decided to run a music competition with 2 categories they are

1. GRAPHICS plus MUSIC
2. MUSIC.

In category 1 the prize money is $40.00 for any music and graphics program.

In category 2 the prize money is $25.00 for any music program.

RULES

1. You can make your program as big or as small as you like.
2. Programs will be judged on originality and ideas.
3. The programs will be judged by COMPUTER FORUM staff.
4. You may send in as many entries as you like.
5. All decisions will be final.
6. All entries will be published in COMPUTER FORUM in the way of listings.

REQUIREMENTS

All you have to do is write up your program, put it on a tape or disk, we recommend you put it on at least twice in case of a bad copy.
If you want your tape back please **include a stamp self addressed post pak** or sufficient funds for return postage.

If you are sending us an entry from outside W.A. please mark the package DO NOT XRAY - COMPUTER TAPE. I don't know if the xrays do affect them but I would not take any chances.

Please include the name of your program when sending it in.

Include your NAME, ADDRESS and PHONE NUMBER when sending in your entry, so we may contact you.

Send all entries to MUSIC COMPETITION C/- our post office box.

HELP

Here is a little help for all the uses who don't know about music on your computer.

```
10 A$="CDEFGAB"
20 FOR X=1 TO 8
30 PLAY "O=I;XA$;"
40 NEXT I
```

Look in your book for all the music commands associated with PLAY.

MSX BASIC TUTORIAL

First off let me answer a few questions that were sent into me by some readers.

1. What is the difference between CLOAD and BLOAD

CLOAD will only load BASIC programs from tape.

BLOAD will load and run machine programs for EG-

BLOAD"GAME",R

If you leave off the R at the end it WILL NOT run.

So please check for the correct method of loading.

2. Do Commmadooor programs work on the MSX.

NO they do not, you must buy the programs with the MSX symbol on them.

3. How long will it take me to learn how to program.

That is a very good question but a bit hard to answer so let me say roughly November 19th at 7.38pm.

Well I hope if anyone else was going to ask me those questions I have satisfied you.
If you have any questions at all please drop me a line and I will try to do my best to answer them in each issue, if you are in a hurry for an answer, enclose a stamped self addressed envelope and I will write back to you.

Here we go with PART-2 of our TUTORIAL.

In PART-1 we learnt of a few commands and how they worked, in this issue we will expand on those commands plus look at some others, such as -

1.. LIST	4.. RENUM	7.. INKEY$
2.. LIST.	5.. FUNC/KEYS	8.. GOTO
3.. AUTO	6.. KEY LIST	9.. GOSUB

Let me explain what numbers 1 to 6 can do for you to be helpful in programming.

1.. LIST This function speeks for itself, when used it lists a BASIC program onto the screen. There are 2 ways to use this function, 1 type in LIST and press [ENTER] or press FUNCTION KEY [F 4] and [ENTER].

2.. LIST. This has a similar use to the LIST function except when used it lists the last line that has been listed or worked on, press FUNCTION KEY [F 9] to use this option.

3.. AUTO This is a very handy option to use if you use it properly. If you type AUTO or press [F 2] and [ENTER] it will automatically do line numbering for you. Practise using this option to see what it can do for you, it will amaze you.

4.. RENUM This function will renumber you program in lines of 10 or if you specify a number it will number them as that. For example - RENUM 10,100 and press [ENTER] This will renumber all lines starting from 10 and going in steps of 100.

5.. FUNCTION KEYS
The FUNCTION keys are very helpful because you can program them for various tasks. If you wish to type in a lot of things that use the same word you can set up one of your FUNCTION keys for that purpose.

6.. KEY LIST
If you type in the words KEYLIST and press ENTER it will list out the current FUNCTION key settings.

7.. GOTO This function is used in programming to GO TO a particular line.

8.. GOSUB This is similar to the GOTO function except it needs what is called a RETURN statement.

Now lets make up a small program wher you have to guess the missing number this program will use some of the above commands as well as some of the commands we had in part1.

```
10  CLS:KEY OFF
20  RDRST=RND(-TIME)
30  A=INT(RND(1)*500)+1
40  LOCATE 5,2:?"PICK A NUMBER BETWEEN 1-500"
50  LOCATE 10,11:INPUT"YOUR GUESS IS:";G
60  NG=NG+1:LOCATE 10,19:?"No OF GUESSES ";NG
70  IF G<A THEN GOSUB 120
80  IF G>A THEN GOSUB 140
90  IF G=A THEN 160
100 LOCATE 26,11:?"    ":G=0
110 GOTO 50
120 LOCATE 10,15:?"TO LOW!!! TRY AGAIN"
130 FOR X=1 TO 400:NEXT X:LOCATE 10,15:? SPC(19):RETURN
140 LOCATE 10,7:?"TO HIGH!!! TRY AGAIN"
150 FOR X=1 TO 400:NEXT X:LOCATE 10,7:? SPC(19):RETURN
160 LOCATE 2,21:?"DO YOU WANT TO PLAY AGAIN Y or N"
170 A$=INKEY$:IF A$="" THEN 170
180 IF A$="Y" THEN RUN
190 IF A$="N" THEN KEY ON:END
200 GOTO 170
```

If you wish to keep this small program, save it to tape using CSAVE"NUMBER" [ENTER] and don't forget to VERIFY what you have saved CLOAD? [ENTER]

Now lets look at what each command does -

Line 10
KEY OFF - This command turns off the function key display

Line 20
RDRST=RND(-TIME) - This function sets up your random numbers by using increments of time.

Line 30
A - The letter A is set up as a random number between 1 and 500

Line 50
G - Your guess is stored in G

Line 60
NG=NG+1 - The No of guesses you have each time it passes this line NG = 1 more guess.

Line 70
G<A - This means if your 'G'uess is LESS than 'A'nswer then go subroutine at line 120

Line 80
G>A - This means if your 'G'uess is GREATER then 'A'nswer then go subroutine at line 140

Line 90
G=A - If 'G'ues equals 'A'nswer then go to line 160

Line 100
This line sets 'G' back to zero and clears the spot on the screen where the last guess was.

Line 110
Go back to line 50 for another guess

Line 120
Prints message 'To Low !!! Try again"

Line 130
A small time delay loop (so you can read message) then prints blank spaces SPC(19) over the message.

Line 140 - 150
Another message and time loop

Line 160
This prints the question for another game

Line 170
A$=INKEY$ - This line sets a command, and waits for a key to be pressed IN KEY. It means it will do Nothing until a key is pressed.

Line 180
This line asks if A$="Y" then re run the game

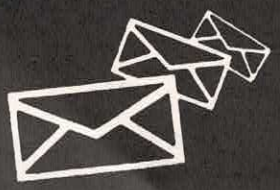
Line 190
This line asks if A$="N" then turn on the FUNCTION key display at the bottom of the screen and END the game.

Line 200
This line acts as a catch all, in other words if A$ does not = "Y" or "N" then GO TO 170. This will happen all the time until A$ = "Y" or "N".

Try changing the number in line 30 (500) to what ever you like and see how many extra guesses it takes you.

Well happy computing and I hope to hear from you.

MAILBOX



Dear Sir,

I am considering the purchase of a disk drive for my MSX (Toshiba HX-10),and a number of questions spring to mind which perhaps you or maybe the readers can answer for me.

1. Are there any drives still on the market compatible with my computer?
2. If not, are secondhand units generally reliable?
3. What configuration do I need to access the full amount of storage space available on a disk?
   (I was thinking of using 3.5" double sided double density disks.)
4. Would there be any problems in using the drives simultaneously with cartridge programs?
5. Any suggestions on which dealers may be able to help me?

Thank you, yours faithfully, Allan Bird, Bomaderry, NSW.

Thank you for your letter, Allan. It is good to see someone still trying to improve and upgrade there system.

As to your questions reguarding disk drives:

1. Yes there are drives still on the market for the MSX range. They are hard to find as there is very few places around still selling hardware and software.

2. The secondhand drives around are generally reliable. The best bet is to put an add in Computer Forum or contact some of the user groups/clubs listed in Computer Forum.

3. To access a double sided 3.5" disk you need to purchase a disk drive with a double sided controller. The only one that I know of was made by Yamaha. Most of the others had a single sided controller.

4. The disk controllers normally occupy one of the cartridge ports or the expansion bus at the rear. There is generally no trouble using cartridge programs as they have a loader program that will either take no notice of the disk drive controller (eg. games) or will allow you to use your disk drive as well (eg. some business application programs).

5. As to dealers, we do not know of many around. You could try-: Greensborough Computers, 67 Grimshaw st.,
   Greensborough Vic. 3088. Ph (03) 434 6166
or contact your state Yamaha distributer, they may be able to tell you where to get a double sided disk drive from.

I hope we have been of some help to you and, if you do find a new disk drive, could you please let us know from whom you purchased it, the price and the number of drives still available.

Dear Ross,

Congratulations on all the work you have put into the last few issues. I realise how little you receive in return.

I am looking to purchase an SV-902 Floppy Disk Drive (For an SV-328) and would appreciate an idea of the price to pay for this and an SV-801 Disk Controller Card (If it is essential).

I have access to an SV-601 Super Expander and I have an SP101 Printer Interface and Star NX-10 printer. I would imagine that, to run them both simultaneously, it would be necessary to put a compatible connector on the sp101 and plug it into the epander with the Disk Drive Controller. Either this or have a "Y" connector with one end going to the 601, and the other going to the SP101.

I was also wondering if you can run normal CP/M programs on the 328 and 902 without modification to them.

The following tip is for the SVI computers and is for backing up any machine code programs on tape.

```
BLOAD"CAS:[name]"
BSAVE"CAS:[name]",&H8000,&HC800,&H8000
```

Yours Sincerely... Joe Maisano. White Hills Victoria

Thank you for the tip. I'm sure there are quite a few SV owners who will make good use of this information. For Disk users, change 'BSAVE"CAS: ' to 'BSAVE"1: ' for drive 1.

The price of SV-901 disk drives, good question. The last time I saw them advertised was on a pamphlet from Greensborough Computers 67 Grimshaw St Greensborough 3088 PH (03)434 6166, and they were on 'SALE' for around $200.00. If you look in our for sale area in the mag, you may be able to get some second hand drives, with software, fairly cheap.

If you do not already have a disk controller, you will need one otherwise you will not be able to use the disk drives. Again, you will probably have to get a second hand SV-801 controller. The alternative idea would be to get an SV-605/605A/605B expander. They have the disk & controller built in, as well as a printer card built in too. There have been a few advertised in earlier Computer Forum mags. Other wise, try writting to:-

SYNODA COMPANY 300 CANAL BANK RD. UNIT 334
VILLE ST. PIERRE, QUE. CANADA H8R 1H3

They have advertised a lot of SVI gear which included what you want. If you do get one of the SV-605 series expanders, I'm sure you could sell the SP101 fairly easily. Let them know where you got their address from.

CP/M. Most CP/M programs require the use of the SV-806 80 Column card. That is to say, you can run the programs ok, but the screen display is not quite the same, as you could quite imagine. The standard screen has a maximum display of 40 columns x 24 lines, trying to fit a display ment for 80 columns x 24 lines on the same screen would mean the 40 column screen would need two lines for every one line on the eighty column screen. A bit of a mess could develop, as well as some confusion when trying to read any text displays on the screen.

As a subscriber to this magazine I would like to tell you out there to write in and help support this magazine not just to sit back and think that someone else will do the job because they probably wont so why not write in if you have any problems or ideas etc, as this is the only magazine we have and I would like to see it stay around for a while as we need it. The reason that I'm concerned is I have watched a user club that had a news letter disappear the same as some others. The club asked for help in questions and support for the news letter and received no response. So get your act together and SUPPORT THIS MAGAZINE . Also there are new titles for MSX coming out all the time:
BOUNDER2 (REBOUNDER),DEATH WISH3, TAI-PAN, HEAD over HEELS, CALIFORNIA GAMES, THE LIVING DAY LIGHTS (JAMES BOND 007), BUBBLER ARMY MOVES, R-TYPE and NEMESIS 2 should I go on.

Craig Hursey.

MODEM USERS

For all you people who are lucky or unlucky enough to have a modem and also have VIATEL hooked up.
I would would like to inform you that there is an MSX bulletin board in the GOLDLINK CLUB.
This is a board supplied to users for messages and discussion, there are a few MSX user's useing this board at present and we would like more. I myself have nothing to do with the GOLDLINK ORGINIZATION.
I am just an interested MSX owner.

For those of you who don't know the direct page number for the MSX board it is *642912# if you aren't a member of GOLDLINK it will cost you 30c to send a message on the board.

If anyone wants to contact me about anything my mailbox number is 795618530#, and by the way get onto them and ask for MSX software I have, the more enquiries the better.

PAGE NUMBERS

GOLDLINK *642#

MSX BOARD *642912#

THE SOFTWARE CLUB *22008#

MY MAILBOX No MB *795618530

BYE FROM NORTH QUEENSLAND

DAVID STEWART

SORT Pt3

by Bruce Goodman

In the first two parts of this series the principles of the Bubble and Insertion sorting techniques were described, and implementations of each method in SV Basic were provided. In this, the final part, two examples of the 'Linked Structures' technique will be discussed, and the final two program segments are included.

To create or build sorted lists using the methods discussed so far has required the shuffling or moving of the actual data items within the array. The Linked Structures techniques do not depend on this data movement. Rather, the data items are stored in the order they are input, and the desired sequence is achieved by the use of pointers or Links. A diagramatic representation of a simple linear linked structure may make this principle clearer.

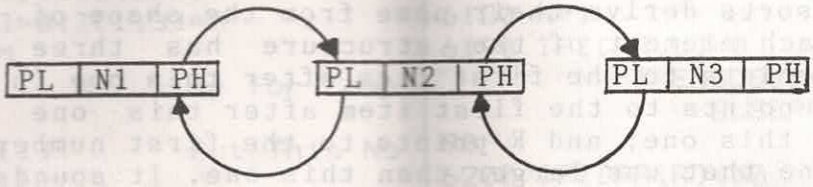


Fig 1

In Fig 1 N1,N2 and N3 are the data entries, and their address in the memory is unimportant. However, stored with each data entry are two pointers, PH (Pointer High) and PL (Pointer Low). These variables contain the actual address of the next sequential item. Now, provided the address of say, the lowest number in the list is known, the entries may be printed in order simply by printing the first item, then using the PH value, locate the next item - and so on. In a similar manner the list could be printed in descending sequence.

Now suppose we wished to insert N4 in the list, such that N2 < N4 < N3, all that is required is to store N4, and re-establish the appropriate links, see Fig 2.

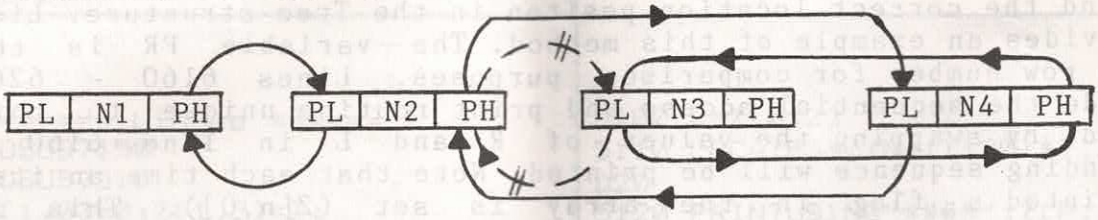


Fig 2

The creation and maintenance of sorted lists by the establishment and amendment of the pointers or links makes this technique ideal for the periodic access and update of sorted lists stored on disk.

List 1 provides a sample implementation of the linear Linked Structures method in SV Basic. Line 5020 sets the initial array conditions. (Z[n,0]=PL, Z[n,1]=the item, and Z[n,3]=PH. Row zero values reserved for pointers to highest and lowest list items.) Line 5040 accepts the new entry, Line 5060 searches through the already established list in sequence until the correct position is found, and the links are amended appropriately. Lines 5100 - 5140 print out the sorted result.

In this Basic implementation, two related factors prevent this method from being noticeably more efficient than previous ones discussed. Firstly, it employs a linear or sequential search for the appropriate position,(similar to the first Insertion method) and, secondly, there are significant computing time overheads in following the pointers around the array.

The 'Tree Sort' method largely overcomes these problems by adopting a 'Binary' data structure shape, see Fig 4.

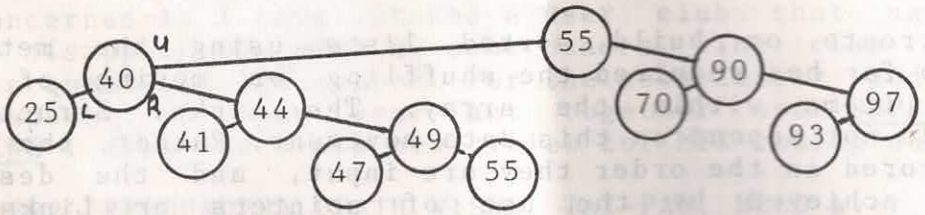


Fig 4

Tree sorts derive their name from the shape of the data structure. Each element of the structure has three links or pointers: U points to the first item after this one in the Tree structure; L points to the first item after this one that was smaller than this one; and R points to the first number entered after this one that was larger than this one. It sounds complex but examination of the example in Fig 4 will aid in gaining an understanding of the concept.

Any new item entered to the list must be located off one of the bottom items already held. Suppose now that another occurrence of the number 55 was entered. It is less than or equal to the first number (55), so we move left, its greater than 40 so move right, greater than 44 so right again. But 49 is on the bottom level in that portion of the Tree, so the new 55 is added to the right of 49.

To print this list in ascending order, all that is required is to print from left to right across the page, descending order from right to left. The efficiency of this method lies in the much smaller number of comparisons required to find the correct location positon in the Tree structure. List 2 provides an example of this method. The variable PR is the array row number for comparison purposes. Lines 6160 - 6260 provide the sequential access and print routine unique to this method. By swapping the values of R and L in line 6160 a descending sequence will be printed. Note that each time an item is printed a flag in the array is set (Z[n,0]). This is necessary as the code does not know whether an item accessed via the U link was accessed from a higher or lower sequential item. This situation may be overcome by the use of an Upper Left and Upper Right link, but the facility of easily reversing the print order by swapping R and L values would be lost.

Summary

This series has attempted to explain the principles, and provide examples of several methods of sorting data items. Each method has its costs and benefits:

Bubble Sorts are simple, require minimum memory and code, but are slow and not suited to disk applications.

Insertion Sorts are quicker, require a little more code and twice the memory to sort an established list. They are also not suited to disk tasks.

Linked Structure Sorts are considerably more complex, (both to understand and code) require their own unique print routines, but the Tree method is very fast by comparison, and ideal for disk use.

Our thanks go to Bruce for the time and effort that he put into this series on Sorting. He has indicated that he is considering doing more articles for us in the near future. As you may have guessed, Bruce has the SV 328 and Disk system, so his articles will be biased towards the SV's. If anyone from the MSX side of things wants to do similar articles, please let us know. All articles are welcome.

==========

```
5990 '========= Tree =========
6000 GOSUB7150
6010 GOSUB7310
6020 Z[1,1]=T[1]:Z[1,2]=0:
       Z[1,4]=0:Z[1,3]=0
6030 FORI=2TONM
6040  PR=1'          Ref line for Comp
are
6050  Z[I,1]=T[I]'       Put This No
 In
6060  IFZ[I,1]<=Z[PR,1]THEN6090
6070  IFZ[PR,4]=0THENZ[PR,4]=I:GOTO
6110
6080  PR=Z[PR,4]:GOTO6060
6090  IFZ[PR,2]<>0THENPR=Z[PR,2]:GO
TO6060
6100  Z[PR,2]=I'                Old L L
ink
6110  Z[I,3]=PR'                New U L
ink
6120  Z[I,2]=0'                 New L L
ink
6130  Z[I,4]=0'                 New R L
ink
6140 NEXT I
6150 GOSUB7290
6160 P=0:L=2:U=3:R=4:PR=1
6170 FORI=1TONM:Z[I,0]=0:NEXTI
6180 N=1
6190 IFZ[PR,L]=0
       THENGOSUB6230
            ELSEPR=Z[PR,L]:GOTO61
90
6200 IFZ[PR,R]<>0
       THENPR=Z[PR,R]:GOTO6190
6210 IFZ[PR,U]=0
       THENIFZ[PR,P]=0
            THENGOSUB6230:RE
TURN
                 ELSERETURN
               ELSEPR=Z[PR,U
]
6220 IFZ[PR,P]=0THENGOSUB6230:GOTO6
200
            ELSE6210
6230 LOCATE3+((N-1)\20)*7,2+(N-1)MO
D20
6240 PRINTUSING"####";Z[PR,1]
6250 Z[PR,0]=1:N=N+1
6260 IFN>100THENRETURN250ELSERETURN
7410 DATA "6..Tree"
```

```
4990 '====== Linked ======
5000 GOSUB7150
5010 GOSUB7310
5020 Z[1,0]=0:Z[1,2]=0:
       Z[0,2]=1:Z[1,1]=T[1]
5030 FORI=2TONM
5040  NO=T[I]'  Get the No
5050  Z[I,1]=NO:P=Z[0,2]
5060  IFNO<=Z[P,1]
        THENZ[I,2]=Z[Z[P,0],2]:
             Z[I,0]=Z[P,0]:
                Z[Z[P,0],2]=
I:
                   Z[P,0]=I
:GOTO5090
5070  IFZ[P,2]<>0THENP=Z[P,2]:GOTO5
060
5080  Z[I,2]=0:Z[P,2]=I:Z[I,0]=P
5090 NEXTI:GOSUB7290:N=1
5100 P=Z[0,2]
5110 LOCATE3+((N-1)\20)*7,2+(N-1)MO
D20
5120 PRINTUSING"####";Z[P,1]:N=N+1
5130 IFN>100THEN5150
5140 IFZ[P,2]<>0THENP=Z[P,2]:GOTO51
10
5150 RETURN
7140 '==== Nos Set Up - Structures
===
7150 DIMZ[NM,4],T[NM]:N=1
7160 FORI=1TONM
7170  Z[I,1]=RND(1)*1000+1
7180  GOSUB7440
7190  T[I]=Z[I,1]:Z[I,1]=0
7200 NEXTI:RETURN
7400 DATA "5..Linked"
```

SVI & MSX SURVEY

Thankyou for taking the time to fill in this survey form, it will help to see how many SVI and MSX machines there are around and how they a used. Please send in this form before 15th MAY 1988. All results will published in Computer Forum as soon as they are available.

Name _____ Address _____

[1] Where did you obtain this survey ?

(A) Computer Forum (B) User Group Newsletter (C) Other

If 'B' Name of user Group _____

If 'C' Name who and where _____

[2] What is your age range ?

(A) 0-10 (B) 11-20 (C) 21-30 (D) 31-40 (E) 41-60 (F) 61-100

[3] What computer(s) do you own ?

(A) SVI-318 (B) SVI-328 (C) MSX (type) (D) MSX 2 (type)
(E) Other (please specify what type)

_____

[4] How big is your computers memory (K) ?

_____

[5] Where did you obtain your computer and why (Please explain).

_____

_____

[6] What type of hardware do you have ?

(A) Cassette (type) _____

(B) Disk Drive (No,type) _____

(C) Printer (type) _____

(D) Modem (type) _____

(E) TV/Monitor (type,BW,etc) _____

(F) Other (Please specify) _____

[7] How do you use your computer ? (Please give % to add up to 100%)

Games ___ % Business ___ % Education ___ % Communication ___ %

Utilities ___ % Robotics ___ % Scientific applications ___ %

Commercial software production ___ % To gain knowledge ___ %

Others ___ % Please explain _____

[8] Do you have any comments about your computer or this survey ?

_____

_____

Please send all forms to -

SVI & MSX SURVEY
C/- JASON WARD, 29 SPENCE STREET
TAREE 2430, N.S.W. AUSTRALIA.

PAYLOAD

(Move over 'SMOKEY', here comes the 'SNAIL', or that's how it seemed to me when I first tried this game. Take my advise, read the directions before you start.)
Think you have the makings of a transport company owner? Test your driving and management skills with 'PAYLOAD'. You'll be rewarded for sound judgment, but you've got to pay for your mistakes!
According to the instruction manual, and I quote "TRAFFIC CONDITIONS IN THE GAME WERE DESIGNED TO BE REALISTIC(in Japan). THERE ARE MORNING AND EVENING RUSH HOURS. AND THERE ARE GOOD DRIVERS AND BAD ONES, SO DRIVE DEFENSIVLEY!"
This means you have to watchout for EVERYONE. Even the Police, who will do a 'U' turn in front of you. So it's back to the workshop for a quick repair job. Meanwhile, you lose time & money even when it's not your fault. I think someone should have words to the insurance people about this. You have to watch which way you move the jostick, or curser keys, while driving. **IF** you successfully complete enough deliveries to accumulate 1,000,000 points (Yen), you too can "BUY THE COMPANY", and be made President of the 'Payload Trucking Company'.

OVERALL COMMENTS -
(Remember these are my comments)

GRAPHICS - EXCELLENT
SOUND EFFECTS - V/GOOD
EXCITEMENT - V/GOOD
VALUE FOR MONEY - EXCELLENT



CHOPLIFTER

Here you are as a pilot of a your helicopter gunship and your mission is to rescue as many of your own men as you can, but you can only carry 16 men at a time so you must fill up your chopper and return to base and come back for the rest, but be careful there is a Tank Corp in the area so you are very vulnerable on the ground and you must avoid them or destroy them.

To make things a bit worse there are also Jet Fighter-Intercepters who try to shoot you out of the sky my suggestion with fighters is to dodge them, you are no match for a Jet Fighter, or are you?

Some of your men that have to be rescued are being held captive in their barracks, you must either fly in low and shoot open the barracks or trick the tanks into doing it by hovering near the barracks.

Tanks can only be destroyed by you flying in very low and shooting them, but to an ace pilot like you this should be no problem.

OVERALL COMMENTS -
(Remember these are my comments)

GRAPHICS - V/GOOD
SOUND EFFECTS - AVERAGE
EXCITEMENT - AVE/GOOD
VALUE FOR MONEY - GOOD



DORODON

This game is based on the old timer "PAC MAN" , although it does have some very distinct differences such as revolving doors windmills, the metamorphic mark and the bonfire. The object of the game is for DORODON to try and avoid the gremlims in the quest to revolve the 99 doors- making them turn the lucky colour of red. You must manover DORODON around the screen avoiding the gremlims, utilise the windmills .(Which if DORODON bumps into them a sudden gust of wind blows, and the gremlims by the red doors will be captured and you will will receive bonus points when you move into the next round.)or the metamorphic mark which when he touches this DORODON gains an extra 10 seconds of strength that enables him to capture the gremlins and increase your score. Then there is the other trap for the gremlins if DORODON is clever enough to lead them into the bonfire and they are destroyed but remembering that if DORODON himself comes in contact with the bonfire he to will be destroyed. If you prove yourself you will receive extra DORODONS at 30,000 and 100,000.

OVERALL COMMENTS -
(Remember these are my comments)

GRAPHICS - V/GOOD
SOUND EFFECTS - GOOD
EXCITEMENT - GOOD
VALUE FOR MONEY - GOOD



SCION

Hostile Aliens occupy the land, and their base is strongly fortified. Destroy them and their base, and end their evil reign.
Quick reflexes are needed to win this one. Be carefull, the game starts straight away, don't wait for the intro music to finish. If you do, then you could lose a man. Capture a Power Unit, this is equipped with bombs.
Destroy all the enemies Energy blocks with your bombs to score and move on to the next scene. There are 3 scenes to each stage. At the end of the third scene, you destroy the Enemy base and move on to the next stage.
IF you clear 99 scenes, your victory is complete.

STRATEGY
Move your fighter around dodging attacks from the enemy, when you see th enemy power unit ship shoot it down and that power unit becomes yours. Once you have the power unit focus in on the Energy Blocks and bomb it. If you destroy he Energy Blocks in numbered order you will receive a bonus TIP for the next scene.

OVERALL COMMENTS -
(Remember these are my comments)

GRAPHICS - V/GOOD
SOUND EFFECTS - GOOD
EXCITEMENT - GOOD
VALUE FOR MONEY - GOOD



>>>>>>>> **MSX** >>>>>>>>>>>>>>>> **CARTRIDGES** <<<<<<<<<<<<<<<< **MSX** <<<<<<<<

Because of the price of the CARTRIDGES we do not hold them in stock, if you wish to buy them please send in your **MONEY and ORDER** to us and we will get them in for you.

Prices include postage in Australia. Overseas add $2.00per Cartridge

TITLE	PRICE
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PRINT LAB	"
MSX HOME WRITER (FOR PRINTER)	$41.00
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MOUSER	"
ALI BARBER & THE 40 THEIVES	"
DORODON	"
CHOPLIFTER	"
LOADRUNNER	"
RAID ON BUNGLING BAY	"
COSMO EXPLORER	"
PAYLOAD	"
ANTARCTIC ADVENTURE	"
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ALPHA SQUADRON	"
SCION	"
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CHARACTER COLLECTION I (Printer Software)	$20.00
CHARACTER COLLECTION II " "	"

CHARACTER COLLECTIONS I & II CAN ONLY BE USED WITH CREATIVE GREATINGS AND ARE PRINTER PLOTTER SOFTWARE.

FOR SALE (Spectravideo Hardware) FOR SALE

2 x SV-902 S/S D/D Disk Drives .....$A120.00 EA
* If both purchased $A200.00
SV-801 Disk Controller Card $A80.00

SV-601 Super Expander $A80.00

SV-802 Printer Card with Cable $A60.00

Special Note The disk drives, if purchased together, come with original CP/M and Disk Basic software as well as programs such as Wordstar, Calastar, dBase2, Datastar etc. and manuals.

Please Contact: Mark Taylor (09) 401 7865

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FOR SALE (MSX Software and Hardware) FOR SALE

Sony 64K HB-75AS, Data Cassette SDC-500
Joystick JS-55 and 3 Cassette games $A160.00

Sony Printer/Plotter PRN-C41 with
Sony Homewriter HBS-H003C cartridge $A300.00
and Printer Cable

Sanyo Light Pen MLP-001 .....$A80.00

MSX-Wizard Robot + Cartridge
MOVIT2-9505 .....$A60.00

7 Asst Cartridge Games $A25.00 EA or the lot $A140.00

UTILITIES

2 Data Cartridges $A35.00 EA or both $A60.00

Music Synthesizer Cartridge $A20.00

OR THE LOT **$A750.00**

contact: The MSX & SVI Users Group of WA
C/- Max Cowin
PO Box 87
Mt Hawthorn
WA 6016

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CP/M Keyboard Map Flag.

by T. Ian Laurie
Gnowangerup W.A.

Under CP/M the keyboard is polled by calling the SV BASIC ROM, thus on return from the ROM, if the keyboard Map Flag is non-zero, then the cursor keys retain their SV BASIC codes. If the flag is 0 then the cursor keys are reassigned as ^E, ^S, ^D or ^X, i.e. WORDSTAR, dBASE etc. type cursor keys.

To change the flag either modify BIOS in memory or BIOS on the system track of the disk. If BIOS in the memory is changed, then this flag will revert to non-zero on each cold boot. Modifying BIOS on the system track provides a permanent fix of this flag. However, you will only need to modify disks that contain programs that require CTRL ESDX type cursor keys.

If disk utility or debugging programs are not available for you to modify this flag, then see CPMCSR.BAS program. This will change the flag on the system track under SV DISK BASIC. It also serves to illustrate how this flag can be changed. Use CPMCSR.BAS on only CP/M 2.20 disks, although CP/M 2.22 disks may be modified if cold boot address at line 360 is changed to &HEEEC. However, any CP/M 2.22 systems I have seen already have this flag set to 0.

Modifying BIOS in memory:-

Flag location in CP/M 2.20 and 2.22 is at &HE660. Change contents of &HE660 to 0 to enable CTRL ESDX type keys. For custom BIOS's then flag will probably be at BIOS + &H60.

Modifying BIOS on system tracks:-

CP/M 2.20 BIOS is stored starting at Track 0 Sector 3 (128 byte sectors) of the system tracks. The keyboard map flag is also located in this sector at byte &H60. Change the contents of this byte to 0 to enable CTRL ESDX type cursor keys.

USER GROUPS

SYDNEY

We have been advised of a group in Sydney for Spectravideo owners- Please contact Malcolm Samuels on 888 7664 after hours.

QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mrs Lucille Parker(Secretary)
25 Primrose St.
Woodridge,Queensland 4114
---- Phone (07)208 5951 ----

VICTORIA

MELBOURNE MSX & SPECTRAVIDEO USERS GROUP INC.
C/- John Halford
48 Shawlands Avenue
Blackburn South Victoria 3130
---- Phone (03)878 6169 ----

We meet on the first Saturday afternoon of each month (except January) at the Nunawading Library meeting room Whitehorse Rd. Nunawading.

GIPPSLAND MSX USER GROUP
C/- Ross Johnson
---- Phone (051)344 753 ----

WESTERN AUSTRALIA

THE MSX & SPECTRAVIDEO USERS GROUP OF W.A
C/- Max Cowin
PO Box 87 Mt Hawthorn
Western Australia 6016
---- Phone (09)417 3323 ----

Meetings are held on the 4th Tuesday of each month at the Colin st. Hall South Perth (Kensington) at 7.30pm.

NEW ZEALAND

WELLINGTON SPECTRAVIDEO USERS GROUP
PO Box 26-050
Newlands, Wellington
New Zealand 6004

CANADA

SMUG of Montrel
20 - 159th Avenue
N.D. Ile Perrot, Quebec
J7V 5V6 Canada

FINLAND

SVI/MSX Club/Turku
C/- Mika Ristolainen
Lumikonkatu 7H 272 SF-20240
Turka Finland

If any of the above groups wish to enter their meeting times & place or phone number, drop us a line and we will include them in this page.

- ASCII CORPORATION -

This is a press release that I received from the ASCII corporation in JAPAN, it may clear up some questions on how good or popular MSX is.

WHAT IS ASCII

In Japan, ASCII means MSX, the most popular home computer in the country. ASCII is Japan's oldest and largest personal computer software, systems, and publications house.
ASCII created the MSX concept in 1983, and holds exclusive worldwide rights to the MSX trademarks, MSX-BASIC, MSX-DOS, and other MSX systems products.

From it's start as a producer of personal computer-related publications, ASCII has diversified to include software and systems development, network services and, more recently, IC chip design and production. Thus, ASCII has evolved from a publisher and software developer to become a Total System Integrator.

ASCII was founded in 1977, and was the Far East representative of Microsoft Corporation, USA, from 1978 to 1986. ASCII recently celebrated it's 10th anniversary as a youthful and vital, yet mature company with nearly 400 employees and steadily growing sales that exceeded US$ 110,000,000 in 1986.

WHO IS ASCII

ASCII is a company of people dedicated to the development and promotion of personal computer software, systems, and services.
ASCII was founded by Kazuhiko (Kay) Nishi and Keiichirou Tsukamoto, college students at the time, and Akio Gunji.
Kay Nishi, age 31 and currently ASCII's president, is well known for his great energy and individual style and articles about him have appeared in Time magazine, the Wall Street Journal, and the Financial Times of London. He lives in a major hotel near ASCII in Tokyo, often travels locally by helicopter, and is said to keep a schedule based on Hawaiian time to help overcome jet lag on his frequent travels.

Less conspicuous, but perhaps even better known indirectly are the creators of MSX -- Jey Suzuki and Rick Yamashita - and the team of MSX developers working on new MSX software and hardware.
Every MSX in the world bears their unique contributions to it's function and style. Though young (the average age at ASCII is 27), they are seasoned professionals at the top in their field.

WHAT IS MSX

MSX is a concept. It is a hardware and software standard accepted by manufacturers and developers in many countries worldwide. The specification was developed to be the ideal machine for home use. Though originally related in some respects to products from Microsoft, the system software was created and developed, and continues to be enhanced, by ASCII.

Key points of the concept are the use of co-processors for powerful graphics, video, and sound capabilities, expandability through external slots, ROM cartridge software, and multi-language capabilities. These features help to keep MSX powerful and flexible, yet simple and inexpensive.

SOME HIGHLIGHTS IN DEVELOPMENT OF MSX AT ASCII

1983

6/83 ---- 14 major Japanese manufacturers, including Sony and Matsushita agree to adopt MSX standard.

9/83 ---- ASCII begins development of MSX-DOS and MSX-DISK BASIC

10/83 ---- MSX announced officially by major manufacturers

1984

5/84 ---- Sony releases 3.5-inch floppy disk drive for MSX

9/84 ---- MSX released in Europe by Phillips and others, release in USSR, Middle East and South America soon follows

1985

5/85 ---- MSX2 introduced, offering enhanced graphics and sound capabilities; MSX2 machines offered by makers

12/85 ---- MSX Festival celebrates sales of 1,000,000 units of MSX. Full-size model of dinosaur mascot constructed in Tokyo

1986

7/86 ---- MSX2 NAPLPS Videotex terminal offered by Sumitomo Trading Company

8/86 ---- Release of MEGA-ROM specification sparks MEGA-ROM wars as software makers rush to create new software

10/86 ---- MSX CLUB organized
Panasonic releases MSX2 priced at Y29,000 (US$195).
ASCII releases MSX-JE, a Japanese language front-end processor for MSX2.
Japan National Broadcasting Corporation announces MSX-based computer correspondence course.

12/86 ---- MSX-NET established, providing network services for MSX users in Japan.

1987

3/87 ---- MSX Group International Section created.
MSX-DOS TOOLS development kit released by ASCII

4/87 ---- MSX-FAN monthly publication begun by Tokuma Shoten Intermedia, Inc.

7/87 ---- MSX-OHENDAN monthly publication begun by Tairiku Shaboh Corporation.

WHO ARE MSX USERS

Because of the flexibility of MSX, there is a wide variety of users of the system. In Japan, as in many other countries, MSX is most popular for recreational applications including games and simulation, music synthesis, composition and performance, and drawing and painting. Over 1,800,000 machines have been sold in Japan to date, mostly in basic models, and over 1000 software titles are available. One hotel in Tokyo offers MSX in each room, networked to a hard disk system that offers a wide selection of games, and can also be used for reservations and room service.

There are, in addition to recreational applications, many other more serious uses for MSX. These include business productivity software, telecommunications, video editing, Videotex, video communications, CD-ROM and laserdisk, fax, and networking. MSX has been selected as a terminal for the Japan Telephone and Telegraph Corporation CAPTAIN videotex system, and is also the basis for a computer correspondence course offered by the Japan National Broadcasting Corporation. An MSX CD-ROM system with several databases has recently been released.

For application developers, there are the MSX-DOS1 and MSX-DOS2 operating systems and a windowing Visual Shell, a BASIC compiler, MSX-DOS Tools, MSX-C Compiler, and the symbolic debugger MSX-SBUG. These packages provide users with an efficient MSX development environment.
In much of Europe also, the largest group of users is interested in recreation. In certain countries, however, wordprocessors, spreadsheets, CAD, and other applications are also popular. An MSX version of dBase II. the popular DBMS, is now on the market, and an MSX-CP/M system has recently been released. There are at least three different networking systems available for MSX that range from a simple hard-disk sharing system to a full featured communications and fileserver system. Networking is especially popular for educational applications, where there is great pressure to keep costs low by sharing resources.

In Argentina and in Arabic countries, MSX is almost a de facto standard, because of its price/performance and flexibility, and is therefore used in small businesses and schools, as well as in the home. The Arabic MSX offers a complete Arabic language environment, and a great deal of business and productivity software, as well as certain special-purpose applications, have been written for that system.

WHAT IS THE FUTURE OF MSX?

The future of MSX is, in a word, bright! A number of new developments promise to bring new dimensions of power and flexibility to the system. MSX graphics, video and sound capabilities will give detail and realism to games and simulations that will rival any other personal system on the market. There will also be more attention focussed on special applications, such as NAPLPS Videotex, telecommunications, video-editing and creativity.

In addition to the A-V advances, such as windowing and large memory management will greatly enhance productivity applications and encourage expanding development of tools to make MSX a powerful business and engineering system. In those areas of the world where MSX is already the standard, these advances will further establish MSX as the leader.

ASCII is committed to a bright and long future for MSX, and to bringing the standard to new markets where price/performance and flexibility are the major requirements.

For further information please contact:

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Three-F Minami Aoyama Bldg.
Minami Aoyama, Minato-ku
Tokyo 107
JAPAN

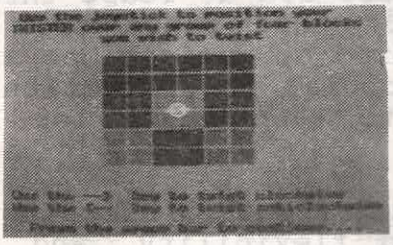
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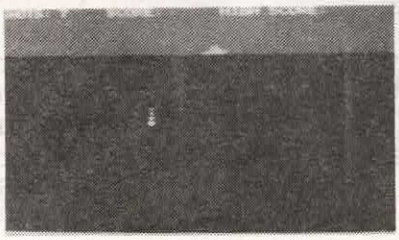
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SVPAK23 - MSXPAK23 -

TWISTER - Like a two dimensional Rubiks cube, this game displays nine different coloured squares each containing four smaller squares. These 36 squares are then jumbled around on the screen and you must rebuild the original display. Its very simple - VERY addictive!

REVERSE - Your task is to arrange the ten jumbled numbers into ascending order by reversing some or all of them. Check it out - its not easy!



TS852 - MSX852 -

GRAND PRIX - You are in the drivers seat in a race to the chequered flag while avoiding slower cars. Hitting another car or two wheels off the track will slow you down and lose points.

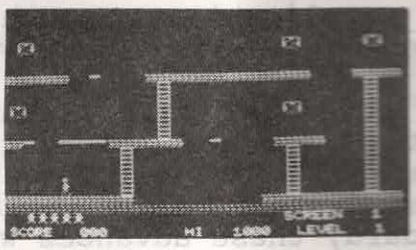
SUBSTRIKE - Drop depth charges on the enemy below. The deeper he is or the faster he moves, the more you will score.



SVPAK24 - MSXPAK24 -

FOX & GEESE - Can you as the Fox penetrate the line of advancing geese and get to the other side of the board before they corner you.

THINK - This is a form of word mastermind. You must guess the word that the computer has selected from the vast number stored in its data banks. This game will certainly make you do as the name suggests.



SVPAK7 - MSXPAK7 -

BOMB SCARE - Very addictive game in which you must climb to different platforms and defuse 5 bombs before your bonus runs out. Has 6 separate screens with the hazards of moving platforms and conveyor belts with holes in them.

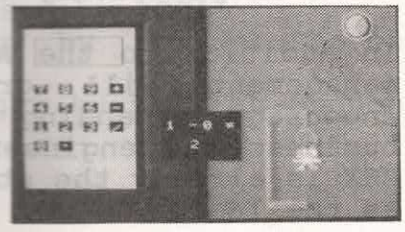
DUNGEON ADVENTURE - A text adventure where you explore the dungeon fighting monsters and gain treasures to discover the location of the great Orb.



SVPAK21 - MSXPAK21 -

GOLDRUSH - Move the miner around the different levels of the mine collecting gold and dodging high explosive charges and falling rocks. The green ledges crumble if you don't get off quickly. When you have collected all of the gold, get to the lift cage for a new screen.

PYRAMID - This is another fascinating text adventure built around the vaults of a pyramid.



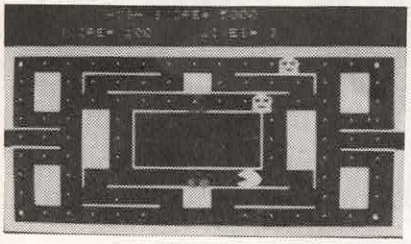
SVPAK25 - MSXPAK25 -

OZMAP - You must drive the car around Australia finding the places required as quickly as you can.

OZQUIZ - Answer general knowledge questions about Australia and her people.

MATH DUEL - Makes practicing maths fun! Has 4 levels of addition and subtraction maths.

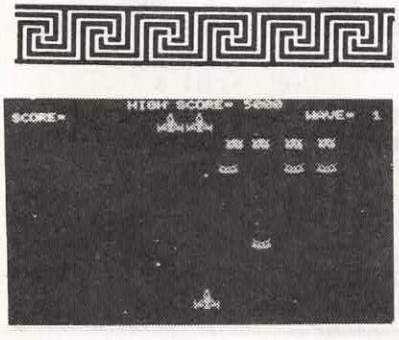
Prices include postage and packing within Australia



SVPAK12 - MSXPAK12 -

MUNCH MAN - Guide your Munchman around the maze eating the dots while avoiding the deadly ghosts. If you eat one of the larger dots you will be able to eat the ghosts for a short while. Gain extra points by eating the fruit that occasionally appears on the screen.

ESCAPE FROM PLANET BETA - This is a text adventure where you are stranded on Planet Beta and must find all the necessary equipment to get off before the planet explodes. Only requires one or two-word commands - eg GET LASER.



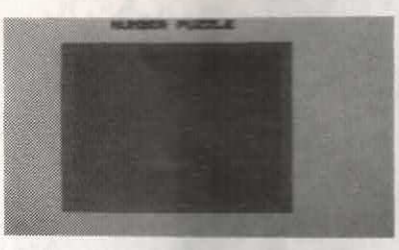
SVPAK5 - MSXPAK5 -

LUNAR LANDER - You are in command of a Lunar Excursion Module with limited fuel but full rotational ability and must land on the lunar surface. Has 8 different landscapes which get harder

GALACTIC ASSAULT - The alien invaders hover above you and one at a time dive on you. Destroy each wave before they get you. For 1 or 2 players.

ROAD RACER - Try and pass as many cars just as fast as you can without crashing.

SPRITE DEFINER - Design sprite shapes on the screen and rotate them when you have finished. Has grids for both sprite sizes.



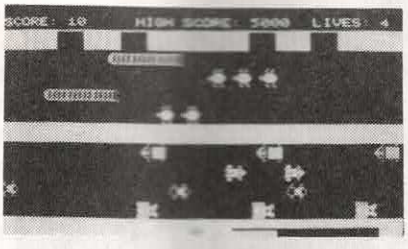
SVPAK20 - MSXPAK20 -

ARTILLERY FIRE - A two player game where each can select a firing angle and velocity to lob shells on his opponent.

NUMBER PUZZLE - Computerised version of the popular hand-held game in which you must slide numbers around into ascending order.

TOWERS OF HANOI - Move all of the discs from peg 1 to peg 3 without placing a large disc on a smaller one.

TENNIS - This is a two player game of Table Tennis.



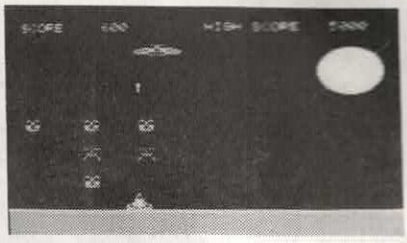
SVPAK6 - MSXPAK6 -

HOPPER - Help Hoppy get across the road (SPLAT!) and then to cross the river (SPLOSH!) and if you are lucky he won't get eaten by the crocs (GULP!!).

SPACE WAR - Challenging game for two players in this duel between your spaceship and that of your opponent. First one to ten points wins.

OTHELLO CHALLENGER - The classic mind game where you must outflank your opponent to capture his pieces. For 1 or 2 players.

PAINTER - Explore the flexibility of your computers colours with this on-screen paint brush.

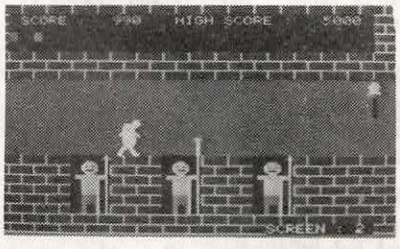


SVPAK9 - MSXPAK9 -

SPACE INVADERS - Shoot the aliens before they invade your planet. Move your ship left and right and shoot missiles at the aliens. Once they have landed the game will end. Score extra points by shooting the UFO which flies overhead.

SUB HUNT - Drop depth charges on the attacking subs and dodge the missiles they fire. You can fire two depth charges at once by firing while moving left or right. Each of the subs is marked with its value.

CHICKEN - Who cares why the chicken crossed the road, we have to find out how! Each time you make it across, the cars speed up.



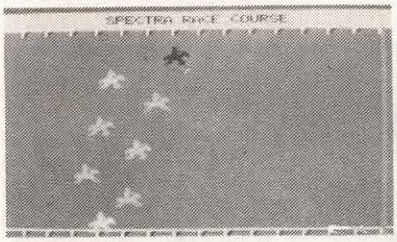
SVPAK11 - MSXPAK11 -

LUNAR RESCUE - Rescue the stranded astronauts from the moons surface one by one. Dodge the asteroids on the way down and destroy them on the way back up for more points.

SKY BASE - Blast the aliens but watch out for their kamikazi attacks. Shoot two sets and then dock with your mother ship. With this added fire-power go on to destroy the alien base.

QUASIMODO - Can you as the hunchback Quasimodo rescue the beautiful maiden from the French soldiers. Cross the screen avoiding fireballs and spears, then ring the bell to get the next screen.

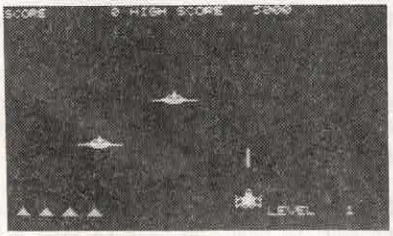
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SVPAK1 - MSXPAK1 -

SPECTRA DERBY - Place your bets and have a day at the races. Even if you lose its painless.

HANGMAN - The classic word guessing game in which you will be gradually hung for each wrong letter.

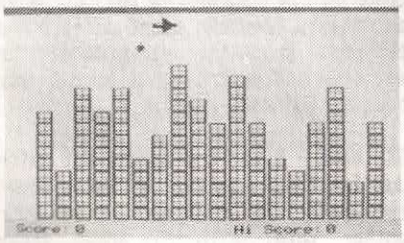


SVPAK10 - MSXPAK10 -

DEMON ATTACK - Shoot the attacking demons as they swoop overhead. You can only destroy a demon when it is fully formed. Killing ten demons gets you on to a harder set.

SEA DIVER - Dodge sea creatures as you make your way to the sea bed. Grab a gold bar and get back to the surface.

DEATH CITY - Race around the maze trying to shoot your computer opponent but be careful, he is also shooting at you.



SVPAK2 - MSXPAK2 -

SNAKE - Move the snake around to eat the diamonds, but mind you do not cross your own tail.

CITY BOMBER - You must destroy the towering buildings of the city before you can land safely.

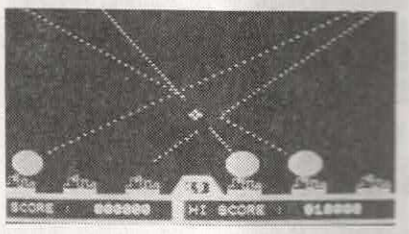


SVPAK8 - MSXPAK8 -

EXPLORER - Watch out for the Pitfalls in this game which will test your control of the joystick. Crocs and other obstacles bar your way as you wander through the jungle in search of treasures.

PARACHUTE - Catch the paratroopers in your boat as they jump from a helicopter into a shark-infested lagoon. The more you catch, the faster they jump.

ANDROID CAVERNS - Move around the caverns of this game and collect the treasure guarded by the androids. But watch out, the androids learn by their mistakes.

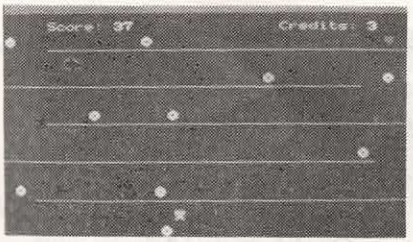


SVPAK22 - MSXPAK22 -

MISSILE COMMAND - Shoot down the incoming warheads before they destroy your cities. You get bonus cities awarded for high scores.

MASTERMIND - Your task is to match the same hidden four-colour combination that the computer has chosen.

LEAKY ROOF - Catch the raindrops leaking through the roof and empty the bucket out of the window before you drown.

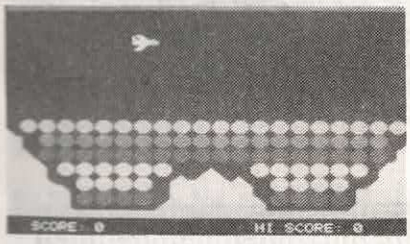


SVPAK4 - MSXPAK4 -

SPECTRA KONG - Win the lady's heart by getting to the top of the screen - but watch out for falling barrels.

TENNIS - A colour version of the original black and white TV game.

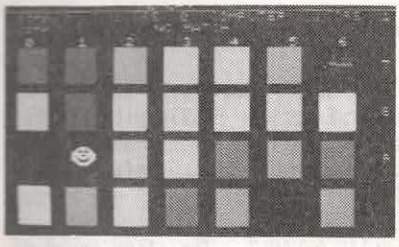
MUSICAL KEYBOARD - This program turns your computer keyboard into a musical keyboard and allows you to select note length and octave.



SVPAK3 - MSXPAK3 -

BOMBS AWAY - Destroy the targets in the ravine below. Be warned - it gets harder the more targets you hit.

BUG CATCHER - Two players move around the garden and the one with the most bugs caught - wins. Great for younger children.



SVEP1 - MSXEP1 -

COPY CAT - The computer will play a set of notes and you must play them back in sequence. There are two levels of play - one note added to the same sequence or a new sequence each time.

MEMORY - A 1 or 2 player game of memory exercise. Uncover two squares on the screen and if the objects revealed match you get another chance

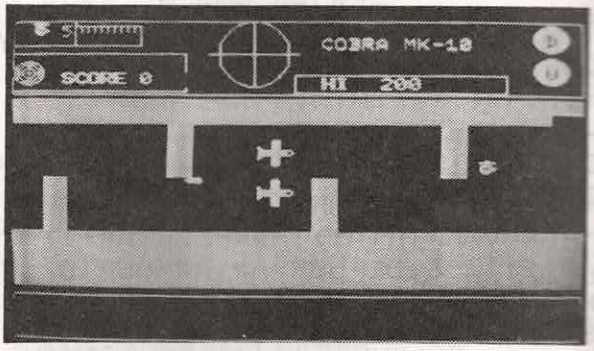
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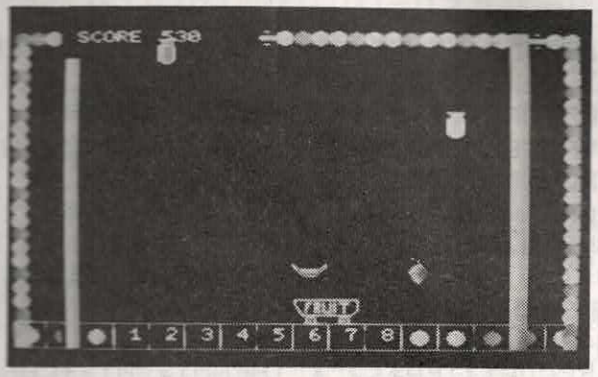
SVPAK26 - MSXPAK26 -

JUMPING COWBOY - A 'platform' game of nine screens where you must collect all the opals to get the next screen.

SUPER SMASHER - Move the bat to smash the ball through the two brick walls. Has three levels of difficulty.

FRUIT CATCHER - Catch falling fruit in your cart. As you catch more, the types of fruit will change and fall at a faster rate.





SVPAK27 - MSXPAK27 -

MIRKWOOD FOREST - Graphics adventure where you must escape from the forest by finding four keys, eating food to maintain energy, while avoiding the spiders. This one will really test your adventure gaming ability.

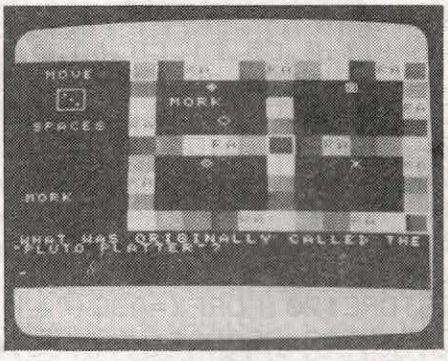
CHOPPER PILOT - The fourteen levels of this game make it very compelling. Keeping clear of canyon walls, fly your helicopter to do battle with enemy fighters, bombers, missiles and UFOs. Great skill required.

UFO (MSX ONLY)

In this machine code game, you are under attack from UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'. Watch out for the meteor shower. Refuel when the truck appears on screen.
Requires 16K RAM, has smooth scrolling, fast response with good sound effects.

METEOR SWARM (SV ONLY)

Another machine code game where your ship is caught in a meteor belt and you must blast your way out. Other ships are caught in the belt and will shoot at you if you do not destroy them first. Comes with bonus game - Birds of Orion.



SPECTRA-TRIVIA (SV 32K)
COMPU-TRIVIA (MSX)

The famous trivia game now on your computer. These are trivia quiz games for up to four players. The computer rolls the dice and each player moves in turn. At certain points you will be asked questions. There is provision for you to create your own question and answer file.

PLEASE SPECIFY SV318/328 OR MSX WITH ORDER.

OTHELLO

for MSX Computers

```
10 DEFINTA-Z:KEYOFF:COLOR15,1,1:SCR
EEN1,2:WIDTH32:CLEAR300:DEFUSR3=&H7
E:RESTORE100:M1$="":M2$="":M3$=""
20 VPOKE&H2004,&HF4:VPOKE&H2005,&H1
2:VPOKE&H2008,&H12:VPOKE&H2009,&H12
:VPOKE&H200A,&H12:VPOKE&H200B,&H12:
VPOKE&H201B,&H21:LOCATE5,2:PRINTSTR
ING$(21,"■"):LOCATE5,3:PRINT"■DANNY
,S■MSX■OTHELLO■":LOCATE5,4:PRINTSTR
ING$(21,"■"):BEEP
30 READA$:IFA$<>"*"THENM1$=M1$+CHR$
(VAL("&h"+A$)):GOTO30
40 LOCATE0,9:PRINTSTRING$(35,"■");"
PLAY■AGAINST■YOUR■MICRO■OR■■■■";
50 READA$:IFA$<>"*"THENM2$=M2$+CHR$
(VAL("&h"+A$)):GOTO50
60 PRINT"■■■■■■AGAINST■A■HUMAN■OPPON
ENT"STRING$(35,"■"):BEEP
70 READA$:IFA$<>"*"THENM3$=M3$+CHR$
(VAL("&h"+A$)):GOTO70
80 LOCATE0,16:PRINTSTRING$(34,"■");
"■LEARN■HOW■TO■WIN■BY■WATCHING■■■■■
■■A■DEMONSTRATION■GAME"STRING$(38,"
■"):BEEP:FORA=1TO8000:NEXT
90 A#=VARPTR(M1$):DEFUSR=PEEK(A#+1)
+PEEK(A#+2)*256:A#=VARPTR(M2$):DEFU
SR1=PEEK(A#+1)+PEEK(A#+2)*256:A#=VA
RPTR(M3$):DEFUSR2=PEEK(A#+1)+PEEK(A
#+2)*256
100 DATA1,0,0,C5,E1,CD,4A,0,11,0,8,
19,CD,4D,0,C5,E1,CD,4A,0,11,0,10,19
,CD,4D,0,3,78,FE,8,20,E2,C9,*,1,1,2
0,21,0,20,CD,4A,0,C5,E1,CD,4D,0,3,7
8,FE,28,20,EF,C9,*,1,0,20,C5,E1,CD,
4A,0,11,0,8,19,CD,4D,0,C5,E1,CD,4A,
0,11,0,10,19,CD,4D,0,3,78,FE,28,20,
E2,C9,*
110 RESTORE120:FORF%=ASC("a")*8TO&H
77+ASC("a")*8:READA$:VPOKEF%,VAL("&
h"+A$):NEXT:CLS:F%=USR(0):F%=USR1(0
)
120 DATA0,0,0,1F,1F,18,18,18,0,0,0,
FF,FF,18,18,18,0,0,0,F8,F8,18,18,18
,18,18,18,1F,1F,18,18,18,18,18,18,F
F,FF,18,18,18,18,18,18,F8,F8,18,18,
18,18,18,18,1F,1F,0,0,0,18,18,18,FF
,FF,0,0,0,18,18,18,F8,F8,0,0,0,18,1
8,18,18,18,18,18,18
130 DATA0,0,0,FF,FF,0,0,0,3C,7E,FF,
FF,FF,FF,7E,3C,3C,7E,FF,FF,FF,FF,7E
,3C,BD,42,99,A5,A5,99,42,BD,3C,7E,F
F,FF,FF,FF,7E,3C:FORF%=&H2000+ASC("
a")*8TO&H2077+ASC("a")*8:READA$:VPO
KEF%,VAL("&h"+A$):NEXT:F%=USR2(0):F
%=USR3(0)
140 DATA21,21,21,21,21,21,21,21,21,
21,21,21,21,21,21,21,21,21,21,21,21
,21,21,21,21,21,21,21,21,21,21,21,2
1,21,21,21,21,21,21,21,21,21,21,21,
21,21,21,21,21,21,21,21,21,21,21,21
,21,21,21,21,21,21,21,21,21,21,21,2
1,21,21,21,21,21,21,21
150 DATA21,21,21,21,21,21,21,21,21,
21,21,21,21,21,F1,F1,F1,F1,F1,F1,F1
,F1,51,51,51,51,51,51,51,51,D1,D1,D
1,D1,D1,D1,D1,D1,D1,D1,D1,D1,D1,D1,
D1,D1:S$="":FORF%=0TO31:READA$:S$=S
$+CHR$(VAL("&H"+A$)):NEXT:SPRITE$(0
)=S$
160 DATA0,0,7,F,14,3A,35,32,32,35,3
A,14,F,7,0,0,0,0,E0,F0,28,5C,AC,4C,
4C,AC,5C,28,F0,E0,0,0:DIMB%(9,9),P%
(60),DX%(8),DY%(8),S1%(2),C$(2),P$(
2),N$(2),QQ%(2):FORF%=1TO8:READDX%(
F%):NEXT:FORF%=1TO8:READDY%(F%):NEX
T
170 DATA0,1,1,1,0,-1,-1,-1,-1,-1,0,
1,1,1,0,-1:C$(0)="  ":C$(1)="1":C$(2
)="m":A$=">"+CHR$(0)+"■2"+CHR$(1)+"
▌":A#=VARPTR(A$):A#=PEEK(A#+1)+PEEK
(A#+2)*256:DEFUSR=A#:A#=USR(0):POKE
&HFCAB,255
180 B%(4,5)=2:B%(5,4)=2:B%(5,5)=1:B
%(4,4)=1:S%(1)=2:S%(2)=2:CP%=1:CO%=
2:FX%=1:FY%=1:RESTORE510:FORX%=1TO6
0:READY%:P%(X%)=Y%:NEXT
190 CLS:PRINT" a";STRING$(28,107);"
c":FORF%=1TO22:LOCATE1,F%:PRINT"j":
LOCATE30,F%:PRINT"j":NEXT:LOCATE1,2
3:PRINT"g";STRING$(28,107);"i";:LOC
ATE6,2:PRINT"DAN JACKSON  OTHELLO":
LOCATE6,3:PRINTSTRING$(20,107)
200 LOCATE3,5:PRINT"a";STRING$(24,1
07);"c":LOCATE1,6:PRINT "dkfHOW MAN
Y PLAYERS (0-2) ?dkf":LOCATE3,7:PRI
NT"g";STRING$(24,107);"i":LOCATE3,1
1:PRINT"(0:DEMONSTRATION GAME)"
210 A$=INKEY$:IFA$=""THEN210
220 IFA$<"0"ORA$>"2"ORLEN(A$)>1THEN
210ELSEFW%=VAL(A$):P$(2)="C":N$(2)=
"COMPUTER":IFFW%=0THENP$(1)="C":N$(
1)="COMPUTER":GOTO260
230 LOCATE3,11:PRINTSPC(27):FORSP%=
1TOFW%:LOCATE2,10:PRINT"NAME OF PLA
YER";STR$(SP%);":":VY%=10:GOSUB590:
N$(SP%)=V$:P$(SP%)="H"
240 A$="PLAYER"+STR$(SP%)+":"+V$:LO
CATE2,10:PRINTSPC(28):LOCATE(28-LEN
(A$))/2+2:PRINTA$:LOCATE3,13:PRINT"
WHICH DO YOU WANT TO USE ?":LOCATE9
,15:PRINT"1:JOYSTICK #1":LOCATE9,16
:PRINT"2:JOYSTICK #2":LOCATE9,17:PR
INT"3:KEYBOARD"
```

```
250 A$=INKEY$:IFA$<"1"ORA$>"3"THEN2
50ELSEQQ%(SP%)=VAL(A$)+3*(A$="3"):F
ORS1%=13TO17:LOCATE3,S1%:PRINTSPC(2
7):NEXTS1%,SP%
260 CLS:PRINT" a";STRING$(28,107);"
c":FORF%=1TO22:LOCATE1,F%:PRINT"j":
LOCATE30,F%:PRINT",j":NEXT:LOCATE1,2
3:PRINT"g";STRING$(28,107);"i";:LOC
ATE3,2:PRINT"n MSX OTHELLO n";TAB(2
2);"SCORE":LOCATE2,4:PRINT"akbkbkbk
bkbkbkbkc":FORY%=1TO8:PRINT" jj";
270 FORX%=1TO8:PRINTC$(B%(X%,Y%));"
j";:NEXT:PRINT:PRINT" jdkekekekekek
ekekf":NEXT:LOCATE2,20:PRINT"gkhkhk
hkhkhkhkhkhki":LOCATE21,5:PRINT" 1 :"
;S%(1):LOCATE21,7:PRINT" m :";S%(2)
:IFFW%>0THENLOCATE3,22:PRINT"PRESS
P TO PASS, H FOR HELP"
280 CP%=1:CO%=2:LOCATE20,10:PRINT"P
LAYER:";C$(1):LOCATE20,12:PRINTSPC(
10):LOCATE20,12:PRINTN$(1):IFP$(1)=
"H"THENGOSUB330ELSEGOSUB420
290 IF(TC%=0ANDNG%=1)OR(S%(1)+S%(2)
=64)ORS%(1)=0ORS%(2)=0THEN520ELSEIF
TC%=0THENNG%=1ELSENG%=0
300 CP%=2:CO%=1:LOCATE20,10:PRINT"P
LAYER:";C$(2):LOCATE20,12:PRINTSPC(
10):LOCATE20,12:PRINTN$(2):IFP$(2)=
"H"THENGOSUB330ELSEGOSUB420
310 IF(TC%=0ANDNG%=1)OR(S%(1)+S%(2)
=64)ORS%(1)=0ORS%(2)=0THEN520ELSEIF
TC%=0THENNG%=1ELSENG%=0
320 GOTO280
330 IFINKEY$<>""THEN330ELSETC%=0:PU
TSPRITE0,(FX%*16+4,FY%*16+19),6
340 PUTSPRITE0,,6:FORF%=0TO30:GO%=S
TRIG(QQ%(CP%)):MO%=STICK(QQ%(CP%)):
A$=INKEY$:IFGO%=0ANDMO%=0ANDA$=""TH
ENNEXTELSE360
350 PUTSPRITE0,,0:FORF%=0TO30:GO%=S
TRIG(QQ%(CP%)):MO%=STICK(QQ%(CP%)):
A$=INKEY$:IFGO%=0ANDMO%=0ANDA$=""TH
ENNEXT:GOTO340
360 IFMO%=0THEN380
370 FORF%=1TO16STEP2:PUTSPRITE0,(FX
%*16+4+F%*DX%(MO%),FY%*16+19+F%*DY%
(MO%)),6:NEXT:BEEP:FX%=FX%+DX%(MO%)
:FY%=FY%+DY%(MO%):FX%=FX%+8*(FX%>8)
-8*(FX%<1):FY%=FY%+8*(FY%>8)-8*(FY%
<1):PUTSPRITE0,(FX%*16+4,FY%*16+19)
,6:GOTO340
380 IFA$="P"THENPUTSPRITE0,,0:RETUR
NELSEIFA$="H"THENPUTSPRITE0,,0:GOSU
B560:GOTO340ELSEIFGO%=0THEN340
390 PUTSPRITE0,,0:X%=FX%:Y%=FY%:SE%
=0:GOSUB460:IFTC%=0THENLOCATE20,15:
PRINT"INVALID!":PLAY"V15T255S8M1000
4C":FORF%=1TO2000:NEXT:LOCATE20,15:
PRINTSPC(8):GOTO330
400 SE%=1:GOSUB460:S%(CP%)=S%(CP%)+
TC%+1:S%(CO%)=S%(CO%)-TC%:LOCATE25,
5:PRINTS%(1):LOCATE25,7:PRINTS%(2):
FORX%=1TO8:FORY%=1TO8:IFB%(X%,Y%)=C
P%THENLOCATEX%*2+1,Y%*2+3:PRINTC$(C
P%)
410 NEXT:NEXT:RETURN
420 LOCATE20,15:PRINT"THINKING!":SE
%=0:FORF%=1TO60:X%=P%(F%)\10:Y%=INT
(P%(F%)-10*X%):GOSUB460:IFTC%=0THEN
NEXTELSEGOTO440
430 LOCATE20,15:PLAY"V15T32L9M2000S
12C":PRINT"I CAN'T GO":FORF%=1TO200
0:NEXT:LOCATE20,15:PRINTSPC(10):RET
URN
440 SE%=1:GOSUB460:S%(CP%)=S%(CP%)+
TC%+1:S%(CO%)=S%(CO%)-TC%:LOCATE25,
5:PRINTS%(1):LOCATE25,7:PRINTS%(2):
FORX%=1TO8:FORY%=1TO8:IFB%(X%,Y%)=C
P%THENLOCATEX%*2+1,Y%*2+3:PRINTC$(C
P%)
450 NEXT:NEXT:RETURN
460 TC%=0:IFB%(X%,Y%)<>0THENRETURN
470 FORD%=1TO8:FO%=0:FP%=0:C%=0
480 C%=C%+1:CX%=X%+C%*DX%(D%):CY%=Y
%+C%*DY%(D%):A%=B%(CX%,CY%):IFA%=CO
%THENFO%=FO%+1:GOTO480ELSEIFA%=CP%T
HENFP%=-(FO%>0)
490 TC%=TC%+FO%*FP%:IFFP%=1ANDSE%=1
THENFORC%=0TOFO%:CX%=X%+C%*DX%(D%):
CY%=Y%+C%*DY%(D%):B%(CX%,CY%)=CP%:L
OCATECX%*2+1,CY%*2+3:PRINT"o":PLAY"
V15T255M1000L6306S8C":NEXT
500 NEXT:IFSE%=1THENLOCATE19,15:PRI
NTTC%;"COUNTER";CHR$(-83*(TC%>1ANDT
C%<10)):LOCATE22,17:PRINT"TAKEN":FO
RF%=1TO1700:NEXT:LOCATE20,15:PRINTS
PC(10):LOCATE22,17:PRINTSPC(5):RETU
RNELSERETURN
510 DATA11,18,81,88,13,31,38,83,16,
61,68,86,33,36,63,66,14,41,48,84,15
,51,58,85,34,35,43,46,53,56,64,65,2
3,24,25,26,32,37,42,47,52,57,62,67,
73,74,75,76,12,17,21,28,71,78,82,87
,22,27,72,77
520 LOCATE20,10:PRINT"GAME OVER":PL
AY"V15T255S8M1000L3005CDEF":LOCATE2
0,12:PRINTSPC(10):IFS%(1)=S%(2)THEN
LOCATE20,13:PRINT"A DRAW!":GOTO540E
LSELOCATE20,13:IFS%(1)>S%(2)THENPRI
NTN$(1)ELSEPRINTN$(2)
530 LOCATE20,15:PRINT"HAS WON BY":L
OCATE19,17:PRINTABS(S%(1)-S%(2));"P
OINTS"
540 LOCATE3,22:PRINT"  PRESS SPACE
TO RESTART   "
550 IFSTRIG(0)=0THEN550ELSERUN
```

```
560 SE%=0:LOCATE20,15:PRINT"HELP":F
ORX%=1TO8:FORY%=1TO8:GOSUB460:IFTC%
>0THENLOCATEX%*2+1,Y%*2+3:PRINT"?"
570 NEXT:NEXT:LOCATE20,15:PRINT"PRE
SS":LOCATE20,17:PRINT"SPACE"
580 A$=INKEY$:IFA$<>" "THEN580ELSEL
OCATE20,15:PRINTSPC(5):LOCATE20,17:
PRINTSPC(5):FORX%=1TO8:FORY%=1TO8:L
OCATEX%*2+1,Y%*2+3:PRINTC$(B%(X%,Y%
)):NEXT:NEXT:RETURN
590 A$=">"+CHR$(0)+"■2"+CHR$(1)+"▌"
:A#=VARPTR(A$):A#=PEEK(A#+1)+PEEK(A
#+2)*256:DEFUSR=A#:A#=USR(0):POKE&H
FCAB,255
600 V$="":LOCATE19,VY%:PRINTSTRING$
(10,46);:X%=19:Y%=VY%:X1%=X%+10:LOC
ATE19,VY%
610 CH%=-46*(X%<X1%)-32*(X%=X1%):LO
CATEX%,Y%:PRINT"?";CHR$(29);:FORF%=
1TO50:A$=INKEY$:IFA$=""THENNEXTELSE
630
620 LOCATEX%,Y%:PRINTCHR$(CH%);CHR$
(29);:FORF%=1TO50:A$=INKEY$:IFA$=""
THENNEXT:GOTO610ELSE630
630 IFA$=CHR$(13)THENPRINTCHR$(CH%)
;:GOTO660
640 IFA$=CHR$(29)THENIFX%>19THENPRI
NTCHR$(CH%);:X%=X%-1:V$=LEFT$(V$,LE
N(V$)-1):GOTO610ELSEGOTO610
650 IF(X%<29ANDASC(A$)>31AND ASC(A$
)<127)THENPRINTA$;:V$=V$+A$:X%=X%+1
:GOTO610ELSEGOTO610
660 LOCATE19+LEN(V$),VY%:PRINTSPC(1
0-LEN(V$));:RETURN
```

```
10 WE NEED MORE  LISTINGS FOR
20 SVI AND MSX, IF YOU HAVE ANY
30 LISTINGS PLEASE SEND THEM IN
40 ON TAPE OR DISK.
50
60 PLEASE REMEMBER IF YOU WANT
70 YOUR TAPE OR DISK BACK INCLUDE
80 A STAMP SELF ADDRESSED ENVELOPE
90
100 I THOUGHT I MAY HAVE STARTED
110 SOMETHING WITH A CARTOON IN
120 THE LAST ISSUE, BUT ALAS I WAS
130 WRONG, IF YOU WISH TO SEND  IN
140 A CARTOON PLEASE FEEL FREE.
```

I thought I would expand a bit on my Editorial on the Software Swapping idea.

I would like to hear from you to see how you would handle it, because I think it will be a bit of a hassle at first, but with proper organization it will become something we can all profit from.

The way I think we should do it is for everyone to send in a list of all the original software you have as per sample below -

MODE	NAME	PRICE	TYPE
MSX	RETURN TO EDEN	$29.95	ADVENTURE GAME
"	DISK WARRIOR	$24.95	3/D ARCADE GAME
"	HERO	$19.95	GRAPHIC GAME
"	PINBALL	$14.95	" "

The above list would then entitle the owner to swap with anyone else who may have different software at the same price. Please write in and we will see what happens.

```
This MENU program is from Bruce
Goodman and is for SV Computers.
for special CHR's see end of list.

10 CLEAR 1000:GOSUB 930
20 OUT&H88,15:OUT&H8C,255:POKE&HFE3
8,1
30 SCREEN0,0:LOCATE,,0:COLOR7,1:CLS
40 DEFSTRA-H:DEFINTI-Z:GOSUB300:DIM
F[35,1]:FIELD#0,128ASD1,128ASD2:P=0
:YR=12
50 FORI=1TO13:C=DSKI$(1,20,I):D=D1:
GOSUB170:D=D2:GOSUB170:NEXTI
60 IFP=0THENGOSUB310:LOCATE,,1:END
70 IFP=1THENP=0:GOTO90
80 P=3*((P-1)\6)+1
90 GOSUB410:GOSUB430:GOSUB120
100 GOSUB550:GOSUB730:GOTO100
110 '<<<<<< Flashing Message >>>>>>
>>>
120 E="Use Joystick to select Progr
am....":GOTO150
130 E="ASCII File...Attempt Run (Y/
N) ?":GOTO150
140 E="Screen File...Can't Run":GOT
O150
150 BEEP:GOSUB910:ONINTERVAL=50GOSU
B480:INTERVALON:RETURN
160 '<<<<<<< Find & Print Files >>>
>>>
170 FORJ=0TO7:E=MID$(D,(16*J)+1,16)
180 IFLEFT$(E,1)=CHR$(255)THENRETUR
N60
190 IFLEFT$(E,1)=CHR$(0)THEN280
200 Z=ASC(MID$(E,10,1))
210 IFZ=160ORZ=224ORZ=176THENG="#":
GOTO250
220 IFZ=0ORZ=16ORZ=64THENG=" ":GOTO
250
230 IFZ=1ORZ=17ORZ=65THENG="*":GOTO
250
240 G="."
250 F[P,1]=MID$(E,1,6)+G+MID$(E,7,3
)
260 IF(P+3)MOD6=0THENLOCATE0,5:PRIN
TCHR$(27)+"M":YR=YR-1
270 GOSUB410:LOCATEX,Y:PRINTF[P,1]:
P=P+1
280 NEXTJ:RETURN
290 '<<<<< Print Screen Heading >>>
>>>
300 A="DISK MENU":Y1=0:GOTO 320
310 A="This disk is Empty !!":Y1=11
320 A=CHR$(211)+CHR$(210)+" "+A+" "
+CHR$(210)+CHR$(211)
330 X1=20-LEN(A)/2
340 LOCATE X1,Y1
350 PRINT"ƹ";STRING$(LEN(A)-2,"I");
"ƺ"
360 LOCATE X1,Y1+1:PRINT A
370 LOCATE X1,Y1+2
380 PRINT"ij";STRING$(LEN(A)-2,"I");
"6"
390 RETURN
400 '<<<<<<< Using P Find X,Y >>>>>
>>>
410  Y=YR+P\3:X=PMOD3+2+(PMOD3)*12:
RETURN
420 '<<<<<< Turn On File P >>>>>>>>
>>
430 LOCATE X-1,Y
440 PRINT CHR$(27)+"p";"■";
450 PRINT F[P,1];CHR$(27)+"q"
460 RETURN
470 '<<<<<<<<<< Message >>>>>>>>>>>
480  LOCATE 19-LEN(E)/2,21
490 IF Z<>0 THEN 520
500  PRINT CHR$(27)+"p";"■";E;CHR$(
27)+"q";
510  Z=1:RETURN
520 PRINT " ";E;CHR$(27)+"q";
530 Z=0:RETURN
540 '<<<<<<<< STICK RESPOND >>>>>>>
>
550 ST=STICK(0)+STICK(1):IFST=0THEN
710ELSEBEEP
560 INTERVALSTOP:P1=-1
570 IF ST=1 THEN P1=P-3
580 IF ST=3 THEN P1=P+1
590 IF ST=5 THEN P1=P+3
600 IF ST=7 THEN P1=P-1
610 IF P1<0 OR P1>35 THEN 690
620 IF F[P1,1]="" THEN 690
630 GOSUB 410
640 LOCATE X-1,Y
650 PRINT" ";F[P,1]
660 P=P1
670 GOSUB 410
680 FOR I=1 TO 50:NEXT I
690 GOSUB 430
700 INTERVAL ON
710 RETURN
720 '<<<<< Trigger Response >>>>>>
730 TR=STRIG(0) + STRIG(1)
740 IF NOT TR THEN RETURN
750 INTERVAL STOP
760 IF MID$(F[P,1],7,1)="."THENGOSU
B890:RUN"1:"+F[P,1]
770 LOCATE 0,21:PRINT CHR$(27)+"l"
780 IF MID$(F[P,1],7,1)<>" "THEN 83
0
790 BEEP:GOSUB 130
800 B=INKEY$:IFB=""THEN800
810 IFB="Y"THENCLS:MID$(F[P,1],7,1)
="."GOSUB890:RUN"1:"+F[P,1]
820 IFB="N"THENGOSUB120:RETURNELSE8
00
830 IF MID$(F[P,1],7,1)<>"#"THEN870
```

```
840 BEEP:GOSUB140
850 FOR I=1TO8000:NEXT I:BEEP
860 GOSUB 120:RETURN
870 CLS:MID$(F[P,1],7,1)=".":GOSUB8
90:BLOAD"1:"+F[P,1],R
880 '<<<<<<< Clear keyboard >>>>>>>
>>
890 B=INKEY$:IFB=""THENRETURNELSE89
0
900 '<<<<<<< Clear Line 21 >>>>>>>>
>>
910 LOCATE0,21:PRINTCHR$(27)+"l":RE
TURN
920 '>>>>>>>> Begin Module <<<<<<<<
<<
930 ZZ$="START.TST"
940 GOSUB 950:GOTO 1450
950 FORI=1TO10:KEY(I)ON:NEXTI
960 ON KEY GOSUB 1120,1160,1310,970
,1190:RETURN
970 INTERVAL OFF
980 COLOR1,1,1:SCREEN0,0
990 CLS:LOCATE 10,10,0
1000 PRINT "Save as ";ZZ$
1010 LOCATE16,12 : PRINT "Y or N"
1020 INTERVAL ON
1030 ON INTERVAL=25 GOSUB 1080 : Z$
=""
1040 Z$=INKEY$:IF Z$="" THEN 1040
1050 IF Z$="Y"ORZ$="y"THENSAVE"1:"+
ZZ$:     INTERVALOFF:COLOR 10,1:CLS
:RUN
1060 IF Z$="N" OR Z$="n" THEN
          INTERVAL OFF:RUN
1070 BEEP:GOTO 1040
1080 COLOR10:ON INTERVAL=25 GOSUB 1
100
1090 BEEP : RETURN
1100 COLOR1:ON INTERVAL=25 GOSUB 10
80
1110 RETURN
1120 GOSUB 1220
1130 INTERVAL OFF
1140 COLOR11,1,1
1150 SCREEN0,1:LOCATE,,1:LIST -930'
1160 INTERVAL OFF
1170 COLOR1,1,1
1180 SCREEN0,0:LOCATE,,0:RUN
1190 COLOR1,1,1
1200 SCREEN0,1:LOCATE,,1
1210 INTERVALOFF:GOSUB1220:COLOR11:
END
1220 CLOSE:KEY 1,"CLRLIN"+CHR$(21)
1230 KEY 2," AUTO":KEY 3," LIST"
1240 KEY 4,CHR$(21)+"CLS:LIST"+CHR$
(13)
1250 KEY 5,CHR$(21)+" RUN"+CHR$(13)
1260 KEY 6," RUN"+CHR$(34)+"1:"
1270 KEY 7,"SAVE"+CHR$(34)+"1:"
1280 KEY 8,"LOCATE,,1"+CHR$(13)+CHR
$(30)+CHR$(21)+CHR$(30)+CHR$(21)+CH
R$(30)+CHR$(21)
1290 KEY 9,"COLOR 2,1,1"+CHR$(13)
1300 KEY 10," FILES"+CHR$(13):RETUR
N
1310 INTERVAL OFF
1320 COLOR 13,1
1330 SCREEN 0,0
1340 LOCATE10,5,0
1350 PRINT "Total Memory = 21191"
1360 LOCATE11,10:PRINT "Memory Left
 =";
1370 PRINT USING "######";FRE(0)
1380 LOCATE 11,15:PRINT "Memory Use
d =";
1390 PRINT USING "######";21191-FRE
(0)
1400 GOTO 1400
1410 ON ERROR GOTO 1420 : GOTO 1460
1420 COLOR10,1:SCREEN0,1:LOCATE9,9,
1
1430 PRINT "Error No";ERR;"in Line"
;ERL
1440 GOSUB1220:PRINT:LIST.:END
1450 GOSUB1220:STOP ON : ON STOP GO
SUB 1190
1460 RETURN
1470 REM Æ is ASCII 176// ▒ is ASCI
I 209
1480 REM ñ is ASCII 164// ij is ASCI
I 185
1490 REM ó is ASCII 162// ▌ is ASCI
I 200
```

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